

Aaron James Havener

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EDUCATION

Bachelor of Science in Mechanical Engineering
Florida State University, Tallahassee, FL | May 2025

EXPERIENCE

Project Engineer Intern, Virtual Design and Construction Engineer Intern

Whiting-Turner | Milton, FL, Tampa, FL | May 2024-August 2024

- Reviewed engineering drawings (Bluebeam) and documentation in communication with subcontractors to assist in development of a \$50 million project
- Supported site development and construction operations at NAS Whiting Field, contributing to the ongoing execution of an \$18 million flight simulator facility
- Worked with project management to maintain material procurement records, vendors, and schedule
- Coordinated with VDC team to support the development of a \$200 million Moffit cancer research building
- Performed interference and design checks of MEP in CAD (Navisworks) to identify potential build issues

Educational CVT Robot for Haptic Teleoperation

Designed and developed an educational robot demonstrating the use of Continuously Variable Transmissions (CVTs) in haptic robotics for teleoperation. The robot used a pair of steerable wheels mounted on a rotating cylinder to control the motion of a scissor-lift linkage, allowing the end effector to accurately trace programmable two-dimensional shapes.

- Designed and implemented a MATLAB feedback control system derived from the system dynamics to accurately position the end effector throughout its virtual surface while maintaining stable system behavior for arbitrary user-defined trajectories.
- Simultaneously computed two independent wheel steering angles using system constraints and inverse kinematics based on the desired end-effector position and scissor-linkage geometry.
- Derived and tuned controller parameters while accounting for hardware limitations, system dynamics, and customer-defined performance requirements to achieve stable tracking.

Autonomous Nonholonomic Mobile Robot

Programmed an RC vehicle to autonomously navigate between user-defined goal points in a mapped environment while avoiding static obstacles, requiring the integration of perception, planning, and control.

- Implemented a breadth-first search path-planning algorithm with a goal-oriented heuristic to generate collision-free paths using mapped LiDAR data for autonomous waypoint navigation.
- Integrated and validated autonomous waypoint navigation and obstacle avoidance, enabling the robot to complete navigation tasks reliably without requiring remote operator intervention.

Ball-and-Beam Feedback Control System

Developed a MATLAB and Simulink simulation of a nonlinear ball-and-beam system to model, analyze, and validate feedback controller performance. A feedback controller autonomously adjusted the beam's tilt to stabilize the ball at user-defined reference positions.

- Derived the nonlinear equations of motion for the ball-and-beam system and built the controller
- Used root locus analysis to tune proportional and derivative gains, demonstrating the relationship between closed-loop pole locations and transient system response.

Merchandiser

Target | Tallahassee, FL | May 2023 – July 2025

Part-time job to support myself while in college. I unloaded trucks, stocked shelves, and helped customers

TECHNICAL SKILLS

CAD: Creo, SolidWorks

Programming: MATLAB, C, C++, Python

Analysis & Simulation: ANSYS (FEA, CFD), COMSOL Multiphysics, Numerical Methods

Mechanical Engineering: Mechanical Design, Gear Design, Mechanism Analysis, Engine Balancing

Controls & Robotics: Feedback Control Systems, Simulink, Dynamic Systems Modeling, Inverse Kinematics, Path Planning, Autonomous Navigation, Sensor Integration, Actuator Control, Electromechanical System Design, OpenCV, Computer Vision

Manufacturing: FDM 3D Printing, Rapid Prototyping

Relevant Coursework and Topics: Mobile Robotics, A*, Dijkstra, Localization, Design and Analysis of Control Systems, State-Space Modeling, Mechatronics, System Dynamics and Vibrations, Machine Design, Mechanical Systems, GD&T, Tolerance Analysis, DFM, Engine Dynamics, Fatigue & Stress Analysis, Weldment Design